

BHU-NETRA: Satellite Ground Segment & Mission Control Platform

Document 01: Comprehensive Functional & Non-Functional Specifications

1. Executive Summary & Mission Overview

BHU-NETRA is an enterprise-grade, national-scale satellite ground-segment and mission control platform (designed on the architectural principles of ISRO's ISTRAC and SAC systems). The system is architected to manage Telemetry, Tracking, and Command (TT&C) operations and payload data handling for a constellation of **40+ Earth Observation (EO) and Communication Satellites**.

The system handles:

1. **Continuous Orbit Tracking & Mid-Pass Handover** across domestic and international ground stations.
2. **Real-time Telemetry Ingestion & Automated Anomaly Detection** for critical satellite health parameters.
3. **Cryptographically Air-Gapped Multi-Sign-Off Command Authorization** workflow.
4. **High-Throughput Payload Ingestion & Automated Orthorectification** processing tens of terabytes daily.
5. **Dynamic Data Classification & Attribute-Based Access Control (ABAC)** for government vs. commercial subscribers.
6. **Constraint-Based Mission Planning & Pass Scheduling** with emergency priority overrides (e.g., disaster response).

2. Core Functional Requirements Breakdown

2.1 Telemetry, Tracking, and Command (TT&C) & Ground Station Operations

- **Multi-Station Orbit Tracking & Steering:** Real-time antenna steering (Azimuth/Elevation) calculated via Two-Line Element (TLE) / SGP4/SDP4 propagation engines.

- **Seamless Mid-Pass Handover:** Handover control from a setting ground station (e.g., Shadnagar, Telangana) to a rising station (e.g., Port Blair, Andaman & Nicobar Islands or Svalbard, Norway) without dropping link state or command lock.
- **Doppler Shift & Frequency Tuning:** Real-time closed-loop tuning of transmitter/receiver frequencies for S-Band, X-Band, and Ka-Band hardware based on relative radial velocity.
- **Software-Defined Radio (SDR) & Legacy Interface Support:** Native capability to interface with older VITA 49 / analog baseband hardware (CPCI controllers, RS-422/serial) and modern SDR platforms (GNU Radio, USRP X410, Kratos Quantum).

2.2 Satellite Health Telemetry & Anomaly Detection

- **Real-time Frame Demuxing:** Processing CCSDS 131.0-B-3 / 132.0-B-2 CADU (Channel Access Data Unit) frames at high ingestion rates.
- **Engineering Unit Calibration:** De-multiplexing raw bitstreams into physical units:
 - *Thermal:* Battery/Solar Array temperature (°C).
 - *Electrical:* Bus Voltage (V), Current Draw (A), Battery State of Charge (SoC %).
 - *Attitude & Orbit Control System (AOCS):* Reaction wheel RPM, star tracker alignment, quaternions, sun sensors.
 - *Propulsion:* Hydrazine fuel tank pressure (Bar), thruster duty cycles.
- **Automated Anomaly Detection & Alerts:** Multi-tier alert thresholds (Yellow/Warning, Red/Critical) evaluated within <50ms. Automated escalation via PagerDuty, SMS, and Mission Control Console alarms.

2.3 Cryptographic Multi-Sign-Off Command Authorization Workflow

- **Air-Gapped Command Security Enclave:** Hardware Security Module (HSM) protected command assembly.
- **3-Stage Authorization Matrix:**
 1. *Stage 1: Satellite Systems Engineer (Operator)* - Drafts telecommand package.
 2. *Stage 2: Flight Dynamics Officer (FDO)* - Validates trajectory, orbital safety window, and power impact.
 3. *Stage 3: Mission Director (MD)* - Final cryptographic sign-off using Hardware Token / PKI.
- **Telecommand Generation:** Framing according to CCSDS Telecommand 231.0-B-3 with BCH error correction and sequence counters to prevent replay attacks.

2.4 Payload Ingestion, Processing & Geo-Referencing

- **High-Throughput Downlink Ingestion:** Receiving multi-spectral, panchromatic, Synthetic Aperture Radar (SAR), and hyperspectral data downlinked at 1.2 Gbps to 10 Gbps per ground station pass.
- **Automated Level-0 to Level-3 Processing Pipeline:**
 - *Level-0:* Raw demuxed payload frames with checksum verification.
 - *Level-1:* Radiometric calibration and sensor nonlinearity correction.
 - *Level-2:* Geometric correction, DEM orthorectification, and geo-referencing (GeoTIFF / COG format).
 - *Level-3:* Mosaic generation, cloud mask filtering, and change detection indexing.
- **Storage Tiering:** Immediate ingest to NVMe Hot Tier, staging to Warm SAN (5 PB), and long-term archiving to Cold LTO Tape / Object Storage.

2.5 Mission Planning & Pass Scheduling Engine

- **Constrained Optimization Scheduler:** Evaluates orbital pass geometry, ground station availability, power budgets, sensor thermal limits, and onboard storage capacity.
- **Weather & Cloud Cover Integration:** Automated ingest of IMD (India Meteorological Department) and ECMWF weather forecasts to prioritize cloud-free imaging passes.
- **Emergency Override Protocol:** Priority queues for national emergency events (e.g., cyclone, flood monitoring by NDRF), auto-preempting routine optical imaging passes within 15 minutes of trigger.

2.6 Data Classification, Entitlements & Dissemination

- **Security Classification Engine:**
 - *Restricted / Defense:* High-resolution imagery of strategic areas; stored strictly on-prem, encrypted with GOV-Grade HSM keys.
 - *Government Ministries (Agri, Disaster, Forestry):* Full resolution imagery delivered via secure, direct G2G network links (NICNET).
 - *Commercial / Academic:* Open / Paid tier with dynamic spatial downsampling, cloud-masking, or geographic boundary redaction.
- **Cloud-Hosted Dissemination Portal:** Public/subscriber web portal supporting OGC standards (WMS, WMTS, WFS, STAC API) hosted on accredited public cloud, completely isolated from Core C2 networks via unidirectional Data Diodes.

3. Non-Functional Requirements (NFRs)

NFR Category	Metric / Requirement	Architectural Mechanism
Availability	99.999% (Five Nines) for TT&C functions	Multi-region active-active GS controllers, VRRP failover
Latency	<50ms telemetry alert processing; <5s command transmission lock	In-memory time-series processing (TimescaleDB / Redis)
Throughput	50+ Terabytes/day payload ingestion across 40 satellites	Distributed parallel processing with Ray/Apache Spark & NVMe arrays
Security	Air-gapped Command Enclave, FIPS 140-3 Level 3 HSM	Unidirectional Data Diodes, PKI Multi-sign-off
Interoperability	CCSDS 131.0, 132.0, 231.0, VITA 49, OGC STAC	Abstracted GS Adapter Interface Layer
Compliance	CERT-In Security Guidelines, ISRO Mission Standards	Immutable Append-Only Kafka Audit Ledger

4. System Boundaries & Air-Gap Architecture

